Heteroskedasticity

School of Management and Economics
The Chinese University of Hong Kong, Shenzhen

Heteroskedasticity

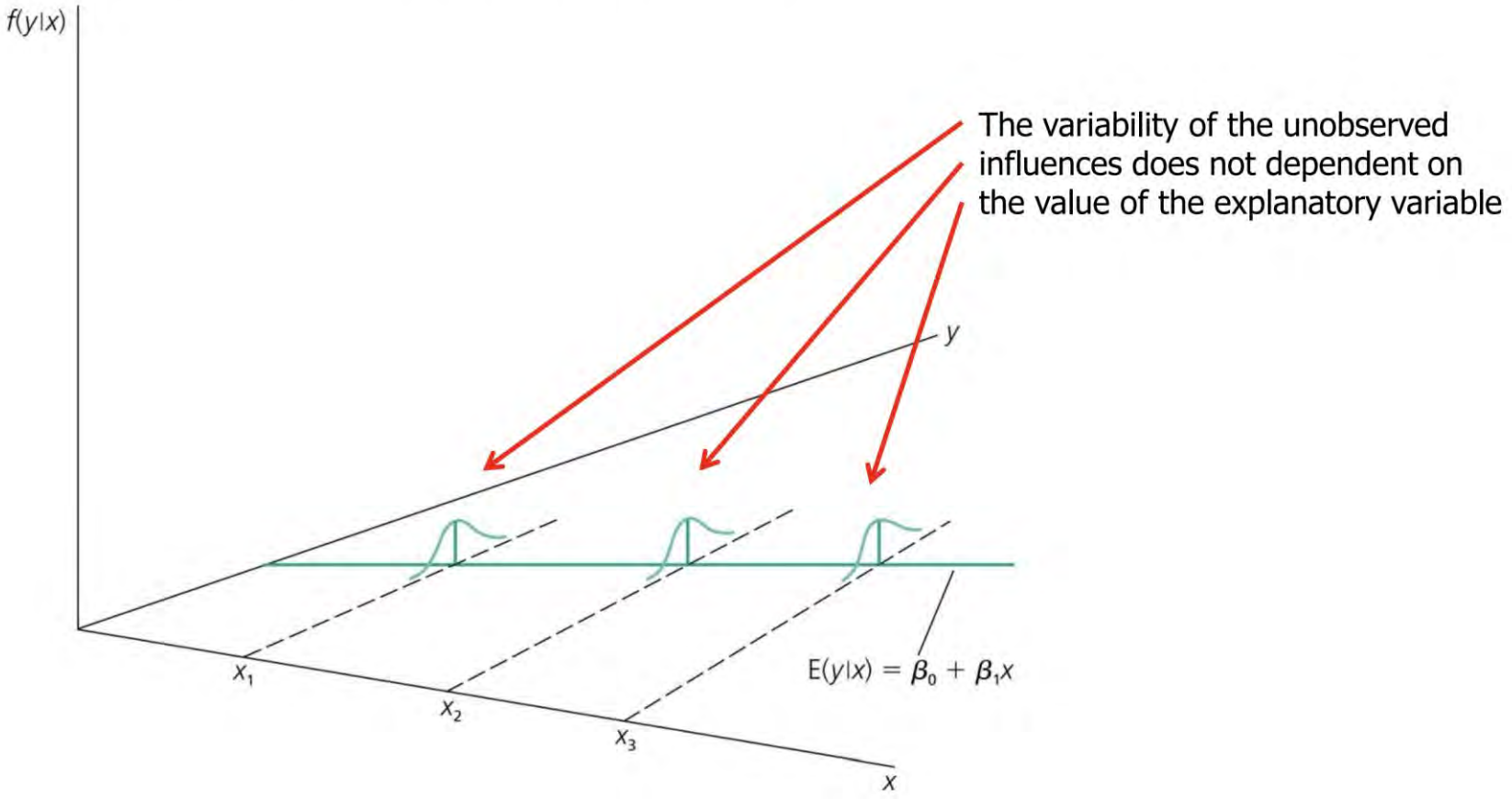
- **Contents**
  - Consequences of Heteroskedasticity for OLS
  - Heteroskedasticity-Robust Inference after OLS Estimation
  - Testing for Heteroskedasticity
  - Weighted least squares estimation

Heteroskedasticity

- **Classical linear model assumptions in MLR:**
  - MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
  - MLR.2: random sampling from the population
  - MLR.3: no perfect collinearity in the sample
  - MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
  - MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)
  - MLR.6: the distribution of u given $(x_1, ..., x_k)$ is $Normal(0, \sigma^2)$

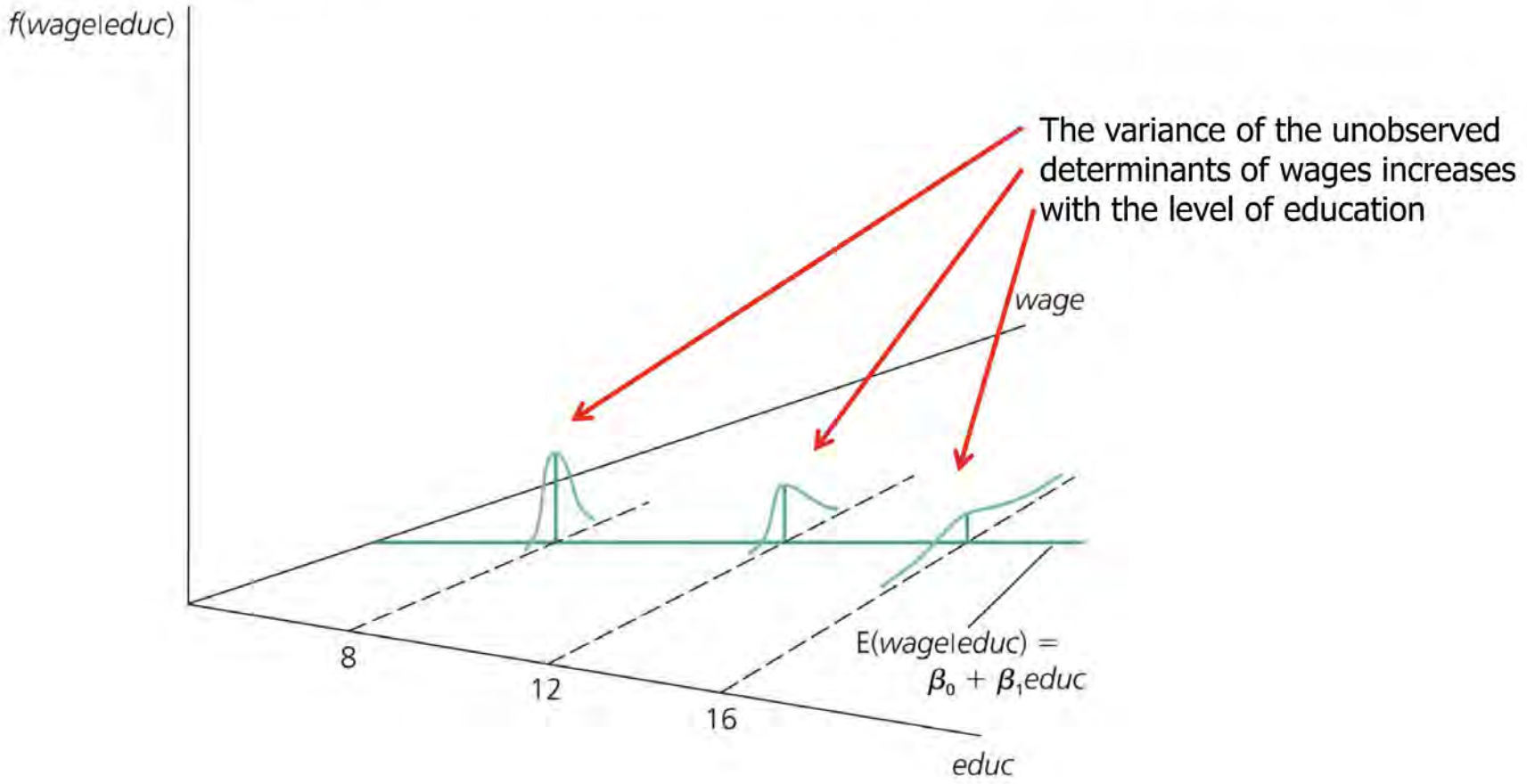
Consequences of Heteroskedasticity for OLS

- **Graphical illustration of homoskedasticity**

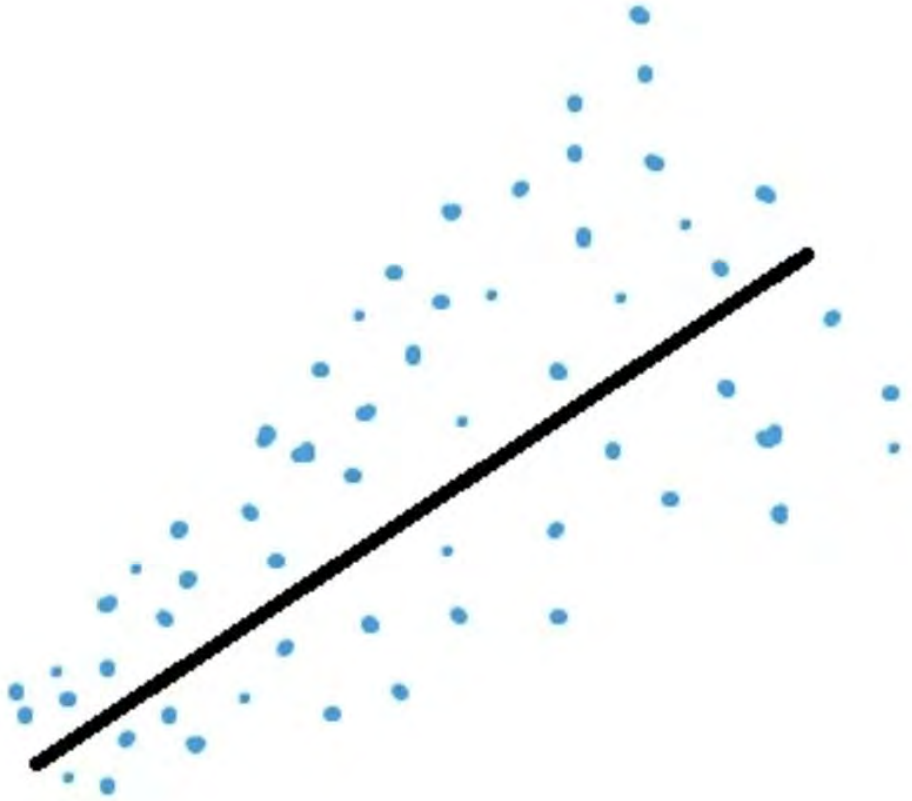


Consequences of Heteroskedasticity for OLS

- **An example for heteroskedasticity: Wage and education**



Consequences of Heteroskedasticity for OLS



Consequences of Heteroskedasticity for OLS

- **Consequences of heteroskedasticity for OLS**
  - OLS still unbiased and consistent under heteroskedastictiy!
  - Also, interpretation of R-squared is not changed
  - Heteroskedasticity invalidates variance formulas for OLS estimators
  - The usual F-tests and t-tests are not valid under heteroskedasticity
  - Under heteroskedasticity, OLS is no longer the best linear unbiased estimator (BLUE); there may be more efficient linear estimators

Heteroskedasticity-Robust Inference after OLS Estimation

- **Heteroskedasticity-robust inference after OLS**
  - Standard errors and all test statistics can be modified to be valid in the presence of **heteroskedasticity of unknown form**
  - Most software include an option with OLS estimation that computes **heteroskedasticity-robust stardard errors**.
  - Formula for heteroskedasticity-robust standard error is not required.
  - In Stata, the general command is

    reg y $x1$ $x2$ ... xk, robust

    where "robust" means "robust to heteroskedasticity of any form."
  - Heteroskedasticity-robust inference are only valid in large samples
  - The usual F-statistic does not work under heteroskedasticity, but heteroskedasticity robust versions are available in most software

Heteroskedasticity-Robust Inference after OLS Estimation

- **Example: Hourly wage equation**

$$\widehat{\log(wage)} = -.128 + .0904\ educ + .0410\ exper - .0007\ exper^2$$
$$(.105)\quad (.0075)\quad (.0052)\quad (.0001)$$
$$[.107]\quad [.0078]\quad [.0050]\quad [.0001]$$

Heteroskedasticity robust standard errors may be larger or smaller than their nonrobust counterparts. The differences are often small in practice.

$H_0 : \beta_{exper} = \beta_{exper^2} = 0$

$F = 42.70$

$F_{robust} = 43.20$

F-statistics are also often not too different.

If there is strong heteroskedasticity, differences may be larger. To be on the safe side, it is advisable to always compute robust standard errors.

Consequences of Heteroskedasticity for OLS

- **Testing for heteroskedasticity**
  - It may still be interesting whether there is heteroskedasticity because then OLS may not be the most efficient linear estimator anymore
- **A few reasons to test for heteroskedasticity include:**
  - We may want to know whether we need to report robust standard errors.
  - We may want to know whether we can improve over OLS, which is possible if there is heteroskedasticity.
  - We may actually want to determine whether the variability changes with the values of the x_j.
- **Useful tests**
  - Breusch-Pagan test
  - White test

Consequences of Heteroskedasticity for OLS

- **Breusch-Pagan test for heteroskedasticity**

$$H_0 : Var(u|x_1, x_2, \ldots, x_k) = Var(u|\mathbf{x}) = \sigma^2$$

$$Var(u|\mathbf{x}) = E(u^2|\mathbf{x}) - [E(u|\mathbf{x})]^2 = E(u^2|\mathbf{x})$$

Under MLR.4

$$\Rightarrow \quad E(u^2|x_1, \ldots, x_k) = E(u^2) = \sigma^2$$

The mean of u^2 must not vary with x_1, x_2, …, x_k

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \cdots + \delta_k x_k + error$$

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_k = 0$$

Regress squared residuals on all explanatory variables and test whether this regression has explanatory power.

$$F = \frac{R^2_{\hat{u}^2}/k}{(1 - R^2_{\hat{u}^2})/(n-k-1)} \sim F_{k,n-k-1}$$

A large test statistic (= a high R-squared) is evidence against the null hypothesis.

Consequences of Heteroskedasticity for OLS

- **Steps in Computing the Breusch-Pagan Test**

  1. Estimate the equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$ by OLS, saving the OLS residuals, $\hat{u}_i$.
  2. Regress $\hat{u}_i^2$ on all explanatory variables and compute the usual F test of joint significance of the explanatory variables.
  3. If the p-value of the test in step 2 is sufficiently small, reject the null of homoskedasticity and conclude Assumption MLR.5 fails.

- **Generic Stata commands:**

```
reg y x1 x2 ...  xk
predict uh, resid
gen uhsq = uh^2
reg uhsq x1 x2 ...  xk
```

Testing for Heteroskedasticity

- **Example: Heteroskedasticity in housing price equations**

$$\widehat{price} = -\underset{(29.48)}{21.77} + \underset{(.0006)}{.0021}\, lotsize + \underset{(.013)}{.123}\, sqrft + \underset{(9.01)}{13.85}\, bdrms$$

$$\Rightarrow\ R_{\hat{u}^2} = .1601, F_{\hat{u}^2} = 5.34,\ p - value_F = .002$$

Heteroskedasticity

$$\widehat{\log(price)} = -\underset{(.65)}{1.30} + \underset{(.038)}{.168}\, \log(lotsize) + \underset{(.093)}{.700}\, \log(sqrft) + \underset{(.028)}{.037}\, bdrms$$

$$\Rightarrow\ R_{\hat{u}^2} = .0480, F_{\hat{u}^2} = 1.41,\ p - value_F = .245$$

In the logarithmic specification, homoskedasticity cannot be rejected

Testing for Heteroskedasticity

- **White test for heteroskedasticity**

Regress squared residuals on all explanatory variables, their squares, and interactions (here: example for k=3)

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2$$

$$+ \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + error$$

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_9 = 0$$

The White test detects more general deviations from heteroskedasticity than the Breusch-Pagan test

- **Disadvantage of this form of the White test**
  - Including all squares and interactions leads to a large number of estimated parameters (e.g. k=6 leads to 27 parameters to be estimated)

Weighted least squares estimation

- **Alternative form of the White test**

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

$$\hat{u}^2 = \delta_0 + \delta_1 \hat{y} + \delta_2 \hat{y}^2 + error$$

This regression indirectly tests the dependence of the squared residuals on the explanatory variables, their squares, and interactions, because the predicted value of y and its square implicitly contain all of these terms.

$$H_0 : \delta_1 = \delta_2 = 0$$

- **Example: Heteroskedasticity in (log) housing price equations**

$$R_{\hat{u}^2} = .0392, F_{\hat{u}^2} = 1.73, \; p - value_F = .1830$$

Homoskedasticity still cannot be rejected, but it has a lower p-value than the B-P test.

Weighted least squares estimation

$$Var\left(\frac{u_i}{\sqrt{h(x_i)}}\middle| x\right) = \sigma^2$$

- **Weighted least squares estimation**
- **Heteroskedasticity is known up to a multiplicative constant**

$$Var(u_i|\mathbf{x}_i) = \sigma^2 h(\mathbf{x}_i),\ \ h(\mathbf{x}_i) = h_i > 0$$

← The functional form of the heteroskedasticity is known

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

$$\Rightarrow \left[\frac{y_i}{\sqrt{h_i}}\right] = \beta_0 \left[\frac{1}{\sqrt{h_i}}\right] + \beta_1 \left[\frac{x_{i1}}{\sqrt{h_i}}\right] + \cdots + \beta_k \left[\frac{x_{ik}}{\sqrt{h_i}}\right] + \left[\frac{u_i}{\sqrt{h_i}}\right]$$

$$\Leftrightarrow\ y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \cdots + \beta_k x_{ik}^* + u_i^*$$

← Transformed model

Weighted least squares estimation

- **Example: Savings and income**



$$sav_i = \beta_0 + \beta_1 inc_i + u_i, \; Var(u_i|inc_i) = \sigma^2 inc_i$$

$$\left[\frac{sav_i}{\sqrt{inc_i}}\right] = \beta_0 \left[\frac{1}{\sqrt{inc_i}}\right] + \beta_1 \left[\frac{inc_i}{\sqrt{inc_i}}\right] + u_i^*$$

Note that this regression model has no intercept

- **The transformed model is homoskedastic**

$$E(u_i^{*2}|\mathbf{x}_i) = E\left[\left(\frac{u_i}{\sqrt{h_i}}\right)^2 |\mathbf{x}_i\right] = \frac{E(u_i^2|\mathbf{x})}{h_i} = \frac{\sigma^2 h_i}{h_i} = \sigma^2$$

- **If the other Gauss-Markov assumptions hold as well, <u>OLS applied to the transformed model</u> is the best linear unbiased estimator!**

Weighted least squares estimation

- **Example: Financial wealth equation** (401KSUBS.dta)

TABLE 8.1 Dependent Variable: *netfa*

Independent Variables	(1) OLS	(2) WLS	(3) OLS	(4) WLS
inc	.821	.787	.771	.740
	(.104)	(.063)	(.100)	(.064)
$(age - 25)^2$	—	—	.0251	.0175
			(.0043)	(.0019)
male	—	—	2.48	1.84
			(2.06)	(1.56)
e401k	—	—	6.89	5.19
			(2.29)	(1.70)
intercept	−10.57	−9.58	−20.98	−16.70
	(2.53)	(1.65)	(3.50)	(1.96)
Observations	2,017	2,017	2,017	2,017
R-squared	.0827	.0709	.1279	.1115

© Cengage Learning, 2013

Net financial wealth

Assumed form of heteroskedasticity:
$Var(u|inc, age, male, e401k) = \sigma^2 inc$

WLS estimates have considerably smaller standard errors (which is line with the expectation that they are more efficient).

Participation in 401K pension plan

Weighted least squares estimation

- **Important special case of heteroskedasticity**
  - If the observations are reported as averages at the city/county/ state/country/firm level, they should be weighted by the size of the unit
- **Example: facters affect pension plan contribution (individual-level)**

Contribution to pension plan for employee e in firm i

Earnings and age for employee e in firm i

Percentage firm contributes to plan

Homoskedasticity error term

$$contrib_{i,e} = \beta_0 + \beta_1 earns_{i,e} + \beta_2 age_{i,e} + \beta_3 mrate_{i,e} + u_{i,e}$$

  - Suppose, however, that we only have firm-level data, i.e., average values of contributions, earnings, and age by firm

Weighted least squares estimation

aggerigate ⇒ 平方差 需要给出权重

- **Firm-level equation**

Average contribution to pension plan in firm i

Average earnings and age in firm i

Percentage firm contributes to plan

Average error across all employees in firm i

$$\overline{contrib}_i = \beta_0 + \beta_1 \overline{earns}_i + \beta_2 \overline{age}_i + \beta_3 mrate_i + \bar{u}_i$$

Let m_i denote the number of employees at each firm

$$\Rightarrow \; Var(\bar{u}_i) = Var\left(\frac{1}{m_i}\sum_{i=1}^{m_i} u_{i,e}\right) = \sigma^2/m_i$$

Heteroskedasticity error at the firm-level (natually)

If errors are homoskedastic at the employee level, WLS with weights equal to firm size m_i should be used.

Feasible Generalized Least Squares

- **Unknown heteroskedasticity function (feasible GLS)**

$$Var(u|\mathbf{x}) = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \cdots + \delta_k x_k) = \sigma^2 h(\mathbf{x})$$

Assumed general form of heteroskedasticity; exp-function is used to ensure positivity

$$u^2 = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \cdots + \delta_k x_k) \cdot v$$

$$\Rightarrow \log(u^2) = \alpha_0 + \delta_1 x_1 + \cdots + \delta_k x_k + e$$

Multiplicative error (assumption: independent of the explanatory variables)

$$\log(\hat{u}^2) = \hat{\alpha}_0 + \hat{\delta}_1 x_1 + \cdots + \hat{\delta}_k x_k + error$$

$$\Rightarrow \hat{h}_i = \exp(\hat{\alpha}_0 + \hat{\delta}_1 x_1 + \cdots + \hat{\delta}_k x_k)$$

Use inverse values of the estimated heteroskedasticity funtion as weights in WLS

Feasible GLS is consistent and asymptotically more efficient than OLS.

Feasible Generalized Least Squares

- **A Feasible GLS Procedure to Correct for Heteroskedasticity:**
  1) Estimate the model by OLS and obtain the residuals, $\widehat{u}_i$.
  2) Create $\log(\widehat{u}_i^2)$ by first squaring the OLS residuals and then taking the natural log.
  3) Run the regression of $\log(\widehat{u}_i^2)$ on all explanatory variables and obtain the fitted values, $\widehat{g}_i$
  4) Exponentiate the fitted values: $\widehat{h}_i = exp(\widehat{g}_i)$
  5) Find WLS estimation using weights $1/\widehat{h}_i$

Feasible Generalized Least Squares

- **Example: Demand for cigarettes** (SMOKE.dta)
- **Estimation by OLS**

Cigarettes smoked per day

Logged income and cigarette price

Smoking restrictions in restaurants

$$\widehat{cigs} = \underset{(24.08)}{-3.64} + \underset{(.728)}{.880}\log(income) - \underset{(5.773)}{.751}\log(cigpric)$$

$$- \underset{(.167)}{.501}\,educ - \underset{(.160)}{.771}\,age - \underset{(.0017)}{.0090}\,age^2 - \underset{(1.11)}{2.83}\,restaurn$$

$$n = 807, R^2 = .0526, p\text{-}value_{Breusch-Pagan} = .000$$

Reject homo-skedasticity

Feasible Generalized Least Squares

- **Estimation by FGLS**

Now statistically significant

$$\widehat{cigs} = \underset{(17.80)}{-5.64} + \underset{(.44)}{1.30}\log(income) - \underset{(4.46)}{2.94}\log(cigpric)$$

$$\underset{(.120)}{-.463}\,educ + \underset{(.097)}{.482}\,age - \underset{(.0009)}{.0056}\,age^2 - \underset{(.80)}{3.46}\,restaurn$$

$$n = 807, R^2 = .1134$$



- **Discussion**

The income elasticity is now statistically significant; other coefficients are also more precisely estimated

The Linear Probability Model Revised

- **WLS in the linear probability model**

$$P(y = 1|\mathbf{x}) = p(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$$

$$\Rightarrow \; Var(y|\mathbf{x}) = p(\mathbf{x})\,[1 - p(\mathbf{x})]$$

In the LPM, the exact form of heteroskedasticity is known

$$\Rightarrow \; \hat{h}_i = \hat{y}_i(1 - \hat{y}_i)$$

Use inverse values as weights in WLS

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

WLS model

$$\Rightarrow \left[\frac{y_i}{\sqrt{h_i}}\right] = \beta_0 \left[\frac{1}{\sqrt{h_i}}\right] + \beta_1 \left[\frac{x_{i1}}{\sqrt{h_i}}\right] + \cdots + \beta_k \left[\frac{x_{ik}}{\sqrt{h_i}}\right] + \left[\frac{u_i}{\sqrt{h_i}}\right]$$

The Linear Probability Model Revised

- **Discussion**
  - Infeasible if LPM predictions are below zero or greater than one (if either $\widehat{y}_i < 0$ or $\widehat{y}_i > 1$, $\widehat{h}_i$ will be negative)
  - If such cases are rare, they may be adjusted to values such as .01/.99
  - Otherwise, it is probably better to use OLS with robust standard errors
- **Estimating the Linear Probability Model by WLS:**

  1) Estimate the model by OLS and obtain the fitted values,
  2) Determine whether all of the fitted values are inside the unit interval. If so, proceed to step (3). If not, some adjustment is needed to bring all fitted values in the unit interval
  3) Calculate $\widehat{h}_i$
  4) Find WLS estimation using weights $1/\widehat{h}_i$

The Linear Probability Model Revised

- **Example: Determinants of Personal Computer Ownership** (GPA1.dta)

Whether the student owns a computer

=1 if at least one parent attended college, 0 otherwise

$$\widehat{PC} = -\underset{(.4905)\ [.4888]}{.0004} + \underset{(.137)\ [.139]}{.065}\, hsGPA + \underset{(.0155)\ [.0158]}{.0006}\, ACT + \underset{(.093)\ [.087]}{.221}\, parcoll$$

$$n = 141, R^2 = .0415$$



WLS estimation

$$\widehat{PC^*} = -\underset{(.477)}{.026}\, 1/\sqrt{h_i} + \underset{(.130)}{.033}\, hsGPA^* + \underset{(.0155)}{.0043}\, ACT^* + \underset{(0.086)}{.215}\, parcoll^*$$

$$n = 141, R^2 = .0464$$